

L1.1 Introduction and Overview Quiz ANS

1. Which autonomous-driving application uses a fleet of robotaxis to provide transportation on demand?

- A) Adaptive Cruise Control (ACC)
- B) Autonomous Mobility-on-Demand (AMoD)
- C) Electronic Stability Control (ESC)
- D) Lane Departure Warning (LDW)

ANS: B) Autonomous Mobility-on-Demand (AMoD)

2. According to the presentation, which driver-assistance feature was initially introduced in 1948?

- A) Cruise control
- B) Surround view
- C) Lane-change assistance
- D) Driver monitoring

ANS: A) Cruise control

3. Which vehicle won the \$2 million top prize in the 2005 DARPA Grand Challenge?

- A) Waymo Firefly
- B) CMU Boss
- C) Stanford Stanley
- D) Tesla Model S

ANS: C) Stanford Stanley

4. Which characteristic is associated with city driving rather than highway driving in the presentation?

- A) High travel speed
- B) Many intersections with traffic lights
- C) Six to eight lanes
- D) Entry and exit points primarily for traffic merging

ANS: B) Many intersections with traffic lights

5. What does an autonomous vehicle's Operational Design Domain (ODD) define?

- A) The vehicle's purchase price and warranty
- B) The neural network's training dataset
- C) The conditions under which the vehicle is designed to operate safely
- D) The driver's preferred route and entertainment settings

ANS: C) The conditions under which the vehicle is designed to operate safely

6. What distinguishes Level 3 conditional automation in the presentation?

- A) It controls only vehicle speed.
- B) It never transfers control to a human.
- C) It can drive without continuous human monitoring but may request that the human take control.
- D) It operates with an unlimited ODD.

ANS: C) It can drive without continuous human monitoring but may request that the human take control.

7. How does Level 4 automation differ from Level 5 automation?

- A) Level 4 controls speed only, while Level 5 controls steering only.
- B) Level 4 operates autonomously under specified conditions, while Level 5 has an unlimited ODD.
- C) Level 4 always requires supervision, while Level 5 never controls the vehicle.
- D) Level 4 applies only to highways, while Level 5 applies only to cities.

ANS: B) Level 4 operates autonomously under specified conditions, while Level 5 has an unlimited ODD.

8. Which statement correctly contrasts the two paths toward Level 4/5 described in the presentation?

- A) Tesla starts with mass-deployed Level 2 and advances upward, while Waymo and nuTonomy start with Level 4 in a limited ODD and expand it.
- B) Tesla begins with Level 5 in a limited ODD, while Waymo begins with Level 1.
- C) Both approaches begin with Level 3 and avoid public deployment.
- D) Waymo begins with mass-deployed Level 2, while Tesla begins with Level 4 robotaxis.

ANS: A) Tesla starts with mass-deployed Level 2 and advances upward, while Waymo and nuTonomy start with Level 4 in a limited ODD and expand it.

9. The May 2016 Tesla fatality was presented primarily as a failure of which capability?

- A) Mission planning
- B) Longitudinal control
- C) Behavior prediction
- D) Computer-vision object detection

ANS: D) Computer-vision object detection

10. The March 2018 Tesla crash into a highway median was associated primarily with a failure of which perception function?

- A) Traffic-light recognition
- B) Pedestrian intention prediction
- C) Lane tracking
- D) Radar-based speed estimation

ANS: C) Lane tracking

11. In California autonomous-vehicle testing reports, what is a disengagement?

- A) An event in which the vehicle hands control to a safety driver or the safety driver takes control
- B) A scheduled shutdown for sensor maintenance
- C) A passenger's decision to cancel a robotaxi ride
- D) A loss of connection between an electric vehicle and its charger

ANS: A) An event in which the vehicle hands control to a safety driver or the safety driver takes control

12. Why total miles driven should not be used as the sole safety metric?

- A) Mileage cannot be recorded by autonomous vehicles.
- B) Only highway mileage is reported.
- C) Companies must drive the same routes and conditions.
- D) Miles can involve very different driving conditions, and companies may avoid difficult conditions.

ANS: D) Miles can involve very different driving conditions, and companies may avoid difficult conditions.

13. Which list contains the four major autonomous-vehicle tasks?

- A) Sensing, mapping, networking, and charging
- B) Detection, braking, steering, and parking
- C) Localization, detection, prediction, and planning/control
- D) Navigation, entertainment, maintenance, and communication

ANS: C) Localization, detection, prediction, and planning/control

14. Which stage of the AV processing pipeline predicts the future movements of detected objects?

- A) Perception
- B) Behavior prediction
- C) Planning
- D) Control

ANS: B) Behavior prediction

15. What is the output of local planning?

- A) A safe and smooth trajectory expressed as vehicle pose over time
- B) A list of raw camera pixels
- C) A destination selected by the passenger
- D) Throttle, brake, and steering actuator signals

ANS: A) A safe and smooth trajectory expressed as vehicle pose over time

16. Which statement describes the end-to-end deep-learning approach to autonomous driving?

- A) It uses a DNN only for static-object detection.
- B) It uses graph search only to select a route on a map.
- C) It outputs a bird's-eye-view representation for separate downstream modules.
- D) It maps input images directly to control and can be trained with imitation or reinforcement learning.

ANS: D) It maps input images directly to control and can be trained with imitation or reinforcement learning.